Residue coverage for the $px + 1$ map: an explicit doubly exponential bound

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Abstract

Let p be an odd prime with $p^2 \nmid 2^{p-1} - 1$, and let $w \geq 1$ with $p \nmid w$ and $w \bmod p$ in the subgroup generated by 2 in $(\mathbb{Z}/p)^\times$. For the map $C(n) = n/2$ (n even), $C(n) = pn + 1$ (n odd), we show that for every $N \geq 2$, every residue r and every B, there is $n > B$ with $n \equiv r \pmod N$ whose C-orbit reaches w, and n can be chosen below $(B + 3)\, w\, 2^{2^F}$ with $F = LN + (3L + 1)\log_2 N + 2$ and $p \leq 2^L$. We also show the condition on $w \bmod p$ is necessary for coverage at $N = p$. The argument is a contraction-and-translation construction on residues, followed by a CRT solve for the exponent; its outer exponent is linear in N, improving a previous polynomial-in-N bound.

1 Setting and statement

Let p be an odd prime and define $C(n) = n/2$ if n is even and $C(n) = pn + 1$ if n is odd. Write $n \rightsquigarrow w$ if $C^k(n) = w$ for some $k \geq 0$. Let $T(n) = n/2$ for even n and $T(n) = (pn + 1)/2$ for odd n. Write $n \rightsquigarrow_T w$ if $T^k(n) = w$ for some $k \geq 0$. Put $H = \langle 2 \rangle \leq (\mathbb{Z}/p)^\times$, and for $m \geq 1$ let $H_m = \langle 2 \rangle \leq (\mathbb{Z}/p^m)^\times$.

Theorem 1. *Let p be an odd prime with $p^2 \nmid 2^{p-1} - 1$, and let $w \geq 1$ with $p \nmid w$ and $w \bmod p \in H$. Let L satisfy $p \leq 2^L$, and let $N \geq 2$, $r \in \mathbb{Z}$, $B \geq 0$. Then there is n with*

$$B < n \leq (B + 3)\, w\, 2^{2^F}, \qquad F = LN + (3L + 1)\log_2 N + 2,$$

such that $n \equiv r \pmod N$ and $n \rightsquigarrow w$.

Theorem 2. *Let p be an odd prime, $p \nmid w$, and $w \bmod p \notin H$. Then the set of n with $n \rightsquigarrow w$ is $\{2^k w : k \geq 0\}$. In particular no $n \equiv 0 \pmod p$ reaches w, so coverage at $N = p$, $r = 0$ fails.*

2 Standard facts used

Only the binomial theorem is used without proof. Everything else is proved below, including CRT for general moduli (Lemma 1) and Fermat–Lagrange for $d = \operatorname{ord}_p 2$.

Lemma 1 (CRT)**.** *Let $g = \gcd(m_1, m_2)$. The system $x \equiv a_i \pmod{m_i}$ is solvable iff $g \mid a_2 - a_1$, and then its solutions form one class modulo $\operatorname{lcm}(m_1, m_2)$.*

Proof. Suppose $a_2 - a_1 = gk$ and $g = sm_1 + tm_2$. Put $x = a_1 + m_1 sk$. Then $x \equiv a_1 \pmod{m_1}$ and $x - a_2 = m_1 sk - gk = -tkm_2 \equiv 0 \pmod{m_2}$. Conversely a solution forces $g \mid a_2 - a_1$. Two solutions differ by a common multiple of m_1, m_2, hence by a multiple of their lcm, and conversely. □

Lemma 2 (Fermat–Lagrange)**.** *Let $d = \operatorname{ord}_p 2$. Then $d \mid p-1$, and $2^{p-1} \equiv 1 \pmod{p}$.*

Proof. 2 is a unit mod p. Lagrange in $(\mathbb{Z}/p)^\times$ (order $p-1$) gives $d \mid p-1$, hence $2^{p-1} = (2^d)^{(p-1)/d} \equiv 1$. □

3 Lifting the exponent

Lemma 3 (binomial step)**.** *Let $j \geq 1$, $x = 1 + p^j b$ with $p \nmid b$. Then $x^p = 1 + p^{j+1} b'$ with $p \nmid b'$.*

Proof. $x^p = 1 + p^{j+1} b + \sum_{i=2}^{p} \binom{p}{i} p^{ij} b^i$. For $2 \leq i \leq p-1$, $p \mid \binom{p}{i}$, so the term has valuation $\geq 1 + 2j \geq j + 2$. The term $i = p$ has valuation $pj \geq 3j \geq j + 2$. Hence $x^p = 1 + p^{j+1}(b + pt)$, and $b' = b + pt$ is prime to p. □

Lemma 4 (valuation identity)**.** *Let $2^d = 1 + pa$ with $p \nmid a$. For $k \geq 1$, $v_p(2^{dk} - 1) = 1 + v_p(k)$.*

Proof. Write $k = p^s k'$ with $p \nmid k'$. Applying Lemma 3 s times gives $(2^d)^{p^s} = 1 + p^{s+1} b_s$, $p \nmid b_s$. Then $(1 + p^{s+1} b_s)^{k'} = 1 + k' p^{s+1} b_s + (\text{terms of valuation} \geq 2s + 2 \geq s + 2)$, so the valuation of the difference from 1 is exactly $s + 1$. □

Corollary 1. *For $m \geq 1$: $2^e \equiv 1 \pmod{p^m}$ iff $d \mid e$ and $p^{m-1} \mid e/d$. Hence $\operatorname{ord}_{p^m} 2 = d\, p^{m-1}$, and $\langle 2^d \rangle = U_1(p^m) := \{x \equiv 1 \pmod{p}\}$.*

Proof. $2^e \equiv 1 \pmod{p}$ forces $d \mid e$; write $e = dk$; Lemma 4 gives $v_p(2^e - 1) = 1 + v_p(k) \geq m$ iff $p^{m-1} \mid k$. Since $|U_1(p^m)| = p^{m-1}$ and $2^d \in U_1(p^m)$ has order p^{m-1}, it generates $U_1(p^m)$. □

Corollary 2 (Lemma 0)**.** *Under the hypothesis $p^2 \nmid 2^{p-1} - 1$, $U_1(p^m) \subseteq H_m$, for every $m \geq 1$. Consequently, if $w \bmod p \in H$ then $w \in H_m$.*

Proof. If $p^2 \mid 2^d - 1$ then $p^2 \mid 2^{p-1} - 1$, since $d \mid p - 1$ (Lemma 2). So $2^d = 1 + pa$ with $p \nmid a$, and Corollary 1 gives $U_1(p^m) = \langle 2^d \rangle \subseteq H_m$. If $w \equiv 2^j \pmod{p}$, then $w 2^{-j} \in U_1(p^m) \subseteq H_m$, so $w \in H_m$. □

4 Two elementary facts about T and C

Lemma 5 (bridge)**.** *If $n \rightsquigarrow_T w$ then $n \rightsquigarrow w$ under C.*

Proof. An even x: $T(x) = x/2 = C(x)$. An odd x: $T(x) = (px + 1)/2 = C(C(x))$, since $C(x) = px + 1$ is even. □

Lemma 6 (affine words)**.** *Let $\omega = a_1 \cdots a_\ell$ be a word in $\{h, u\}$ with o letters u, where $h(x) = x/2$ and $u(x) = (px+1)/2$ are applied in order a_1 first. Define $A_0 = 1$, $C_0 = 0$ and, as letters are read, $(A_{i+1}, C_{i+1}) = (A_i, C_i)$ for h and $(pA_i, pC_i + 2^i)$ for u. Then $A_\ell = p^o$ and $0 \leq C_\ell \leq p^o(2^\ell - 1)$. There is a residue r_ω modulo 2^ℓ such that every $n \equiv r_\omega \pmod{2^\ell}$ has T-orbit with letters $a_1, \ldots, a_\ell$ and*

$$T^\ell(n) = \frac{p^o n + C_\ell}{2^\ell}.$$

Moreover $p^o r_\omega + C_\ell \equiv 0 \pmod{2^\ell}$.

Proof. The formulas follow by induction, since the i-th partial composite is $(A_i n + C_i)/2^i$. For the bound: h keeps C and $2^{i+1} - 1 \geq 2^i - 1$; for u, $pC_i + 2^i \leq p^{o_i+1}(2^i - 1) + 2^i \leq p^{o_i+1}(2^{i+1} - 1)$, the difference being $p^{o_i+1}2^i - 2^i \geq 0$.

Residues, by induction on ℓ. Assume r' works for $\omega' = a_1 \cdots a_{\ell-1}$ modulo $2^{\ell-1}$. For $n \equiv r'$, $y(n) = T^{\ell-1}(n) = (A'n + C')/2^{\ell-1}$ is an integer, and the next letter is h or u according as $y(n)$ is even or odd. Replacing n by $n + 2^{\ell-1}$ changes y by $A' = p^{o'}$, which is odd, so exactly one of the two lifts of r' modulo 2^ℓ gives the required parity. Let r_ω be that lift. Finally $T^\ell(r_\omega)$ is an integer, which gives the congruence. □

5 Contraction and translation on residues

Let $K \geq 0$, $P = p^{K+1}$, $M \geq 1$ with $\gcd(M, 2p) = 1$, and $Q = PM$. On $\mathbb{Z}/Q = \mathbb{Z}/P \times \mathbb{Z}/M$ the maps h, u act componentwise and are bijections.

Lemma 7 (contraction)**.** *Let $x^* = 1/(2-p) \in \mathbb{Z}/P$. Then $u^{K+1}(a) = x^*$ for all $a \in \mathbb{Z}/P$, and $x^* \equiv 1/2 \pmod{p}$.*

Proof. $2-p$ is a unit mod P since $2-p \equiv 2 \pmod{p}$. From $u(x^*) = x^*$ we get $u(a) - x^* = \frac{p}{2}(a - x^*)$, so $u^j(a) - x^* = (p/2)^j(a - x^*)$, which vanishes mod p^{K+1} for $j = K + 1$. □

Lemma 8 (transitivity)**.** *Let $M \geq 1$ with $\gcd(M, 2p) = 1$. The group generated by h, u acting on $\mathbb{Z}/M$ is transitive, and for any s, t there is a word of length at most $M - 1$ taking s to t.*

Proof. $hu(t) = (pt+1)/4$ and $uh(t) = (pt+2)/4 = hu(t)+1/4$. So the translation $\tau_{1/4} = uh \circ (hu)^{-1}$ lies in the group generated by h, u, the latter being a group since it is a finite semigroup of bijections (each generator has finite order, so its inverse is a positive power). As $1/4$ is a unit mod M, all translations lie in the group, so its action is transitive. A shortest word from s to t is a path in the graph on $\mathbb{Z}/M$ and visits distinct vertices, so has length at most $M - 1$. □

6 The word and the exponent

Fix $N = 2^\alpha p^b M$ with $\gcd(M, 2p) = 1$. Let $D = \operatorname{ord}_M 2$, $\nu = v_p(D)$, $d = \operatorname{ord}_p 2$, and take $K = \nu$, $P = p^{K+1}$, $Q = PM$. Note that $p^\nu \leq D \leq M$.

Lemma 9 (word on $\mathbb{Z}/Q$)**.** *Let $z_1 \in \mathbb{Z}$. There is a word W with $\ell = |W|$ and o letters u such that $\ell \leq M - 1 + K + 1 + m$, $m < dp^K$, $o \leq M + K$, and*

$$p^o z_1 + C_W \equiv 2^\ell w \pmod{Q}. \tag{†}$$

Proof. Since $w \bmod p \in H$, Corollary 2 gives $w \in H$ modulo P. Lemma 7 gives $x^* \in H$ modulo P (as $x^* \equiv 1/2 \bmod p$). Choose $m \in [0, \operatorname{ord}_P 2)$ with $2^m w \equiv x^* \pmod{P}$; then $m < dp^K$. By Lemma 8 choose ω of length at most $M - 1$ whose action on $\mathbb{Z}/M$ sends $z_1 \bmod M$ to t^*, where $u^{K+1}(t^*) = 2^m w \pmod{M}$. Let $W = \omega\, u^{K+1}\, h^m$. The P-component ends at $2^{-m}x^* = w$ (Lemma 7 forgets the start) and the M-component at $2^{-m} \cdot 2^m w = w$. Thus W maps z_1 to w modulo Q, which by the affine form of Lemma 6 is (†). □

Lemma 10 (exponent)**.** *Let $c = o_2 + b$, where $o_2 \leq \alpha$ is the number of odd steps in the reduction (Lemma 11). Let $m' = o + c$, $L_{\mathrm{mod}} = \operatorname{lcm}(D, dp^{m'-1})$. Then there is E_0 such that every $E \equiv E_0 \pmod{L_{\mathrm{mod}}}$ satisfies*

$$2^{\ell+E} w \equiv Y := C_W + p^o z_1 \pmod{p^{m'} M}.$$

Consequently $z = (2^{\ell+E}w - C_W)/p^o$ is an integer, $z \equiv z_1 \pmod{p^c M}$, and $T^\ell(z) = 2^E w$.

Proof. Mod M: by (†), $Y \equiv 2^\ell w$, so $D \mid E$ suffices.

Mod $p^{m'}$: by (†), $Y \equiv 2^\ell w \pmod{p^{K+1}}$ (note $K+1 \le o \le m'$). Since $p \nmid 2w$, put $g = Y(2^\ell w)^{-1}$; then $g \equiv 1 \pmod{p^{K+1}}$. By Corollary 2, $g \equiv (2^d)^s \pmod{p^{m'}}$ for some s, determined modulo $p^{m'-1}$. From $g \equiv 1 \pmod{p^{K+1}}$ and $\mathrm{ord}_{p^{K+1}} 2^d = p^K$, we get $p^K \mid s$. By Corollary 1, the condition becomes $E \equiv ds \pmod{dp^{m'-1}}$.

Compatibility. Let $G = \gcd(D, dp^{m'-1})$. The p-part of G is $p^{\min(\nu, m'-1)}$, which divides $p^K \mid s$ since $\nu = K$. For primes $q \neq p$ dividing G, $q \mid d$ and $v_q(G) \le v_q(d) \le v_q(ds)$. Hence $G \mid ds$, and the system $E \equiv 0\ (D)$, $E \equiv ds\ (dp^{m'-1})$ is solvable by Lemma 1, with solutions a class E_0 modulo L_{mod}.

Finally, $p^o z = 2^{\ell+E} w - C_W$ and $p^o z \equiv p^o z_1 \pmod{p^{m'} M}$ give $z \equiv z_1 \pmod{p^c M}$ (since $p^{m'} = p^o p^c$). The congruence $T^\ell(z) = (p^o z + C_W)/2^\ell = 2^E w$ follows from Lemma 6, since $z \equiv r_W \pmod{2^\ell}$ (as $p^o z \equiv -C_W \pmod{2^\ell}$). □

Remark 1. Only $K \ge \nu$ is used. An earlier requirement $K \ge c$ is unnecessary.

7 The 2-power reduction

Lemma 11 (reduction)**.** *Let $r_0 \in [0, 2^\alpha)$ with $r_0 \equiv r \pmod{2^\alpha}$, let o_2 be the number of odd steps among the first α steps of r_0, and $z_0 = T^\alpha(r_0) < 2p^{o_2}$. For $n = r_0 + 2^\alpha m$, $T^\alpha(n) = z_0 + p^{o_2} m$. If $z \equiv z_1 \pmod{p^c M}$ with $z_1 = z_0 + p^{o_2} m_1$ and $m_1 \equiv (r - r_0)/2^\alpha \pmod{p^b M}$, then $m = (z - z_0)/p^{o_2}$ satisfies $n \equiv r \pmod{N}$.*

Proof. The first identity is Lemma 6 with $\ell = \alpha$, since $n \equiv r_0 \pmod{2^\alpha}$. $z_0 < 2p^{o_2}$ follows from $C < p^{o_2} 2^\alpha$. The class statement: $n \equiv r \pmod{2^\alpha}$ holds automatically, and $n \equiv r \pmod{p^b M}$ is $m \equiv m_1 \pmod{p^b M}$, which is equivalent to $z \equiv z_1 \pmod{p^{o_2+b} M}$ because $z - z_1 = p^{o_2}(m - m_1)$. □

8 The bound

Lemma 12 (size of the witness)**.** *Let $S = p^\alpha(B+3)$ and $E^+ = \max(0, \lceil \log_2(p^o(S + 2^\ell)) - \ell \rceil)$. Choose $E \ge E^+$ in the class of Lemma 10, least such. Then $E < E^+ + L_{\mathrm{mod}}$, $z \ge S$, and*

$$z < 2^{L_{\mathrm{mod}}+1}(S + 2^\ell)\, w.$$

Proof. Since $C_W < p^o 2^\ell$, $z \ge S$ holds once $2^{\ell+E} \ge p^o(S + 2^\ell)$, which the choice of E^+ ensures. Since $2^{\ell+E^+} \le 2p^o(S + 2^\ell)$ (using $2^\ell \le p^o(S + 2^\ell)$), we get $2^{\ell+E} < 2^{L_{\mathrm{mod}}} \cdot 2p^o(S + 2^\ell)$, and dividing by p^o gives the bound on z. □

Lemma 13 (exponent estimate)**.** *Let $N \ge 2$ and $p \le 2^L$ (so $L \ge 2$, as $p \ge 3$). Then $\Xi := L_{\mathrm{mod}} + \ell + (L+1)\alpha + 2 \le 2^{\,LN + (3L+1)\log_2 N + 2}$.*

Proof. Since $o \le M + \nu$ and $c \le \alpha + b$, with $M \le N$, $\nu, \alpha, b \le \log_2 N$, we get $m' \le N + 3\log_2 N$. Thus $L_{\mathrm{mod}} < M p^{m'} \le N\, 2^{L(N + 3\log_2 N)} = 2^X$ with $X = LN + (3L+1)\log_2 N$. Also $\ell \le N + \log_2 N + 1 + 2^L N \le 2^{L+2} N$. Since $LN \ge L + N$ for $L, N \ge 2$, we have $2^X \ge 2^{LN} N^{3L+1} \ge 2^{4L+N+1}$, which is at least $2^{L+2} N \ge \ell$ and at least $(L+1)\log_2 N + 2 \ge (L+1)\alpha + 2$. So $\Xi \le 3 \cdot 2^X \le 2^{X+2}$. □

9 Proof of Theorem 1

Let $n = r_0 + 2^\alpha m$ with $m = (z - z_0)/p^{o_2}$, where z is given by Lemma 10 with E from Lemma 12. Then $z \geq S$ gives $m \geq B+1$: indeed $z_0 < 2p^\alpha$ and $p^{o_2} \leq p^\alpha$ give $m \geq (p^\alpha(B+3) - 2p^\alpha)/p^\alpha = B+1$. Hence $n \geq m > B$. By Lemma 11, $n \equiv r \pmod N$. Since $T^\alpha(n) = z$ and $T^\ell(z) = 2^E w$ and E halvings reach w, we have $n \rightsquigarrow_T w$, hence $n \rightsquigarrow w$ by Lemma 5.

For the bound: $n \leq 2^{\alpha+1} z$ (as $r_0 < 2^\alpha$, $z \geq 1$). Lemma 12 gives $z < 2^{L_{\mathrm{mod}}+1}(S + 2^\ell)w$. Using $S + 2^\ell \leq (B+3)p^\alpha 2^\ell$ and $p^\alpha \leq 2^{L\alpha}$,

$$n < (B+3)\, w\, 2^{\Xi}, \qquad \Xi = L_{\mathrm{mod}} + \ell + (L+1)\alpha + 2,$$

and Lemma 13 gives $\Xi \leq 2^F$. Hence $n \leq (B+3)w2^{2^F}$. □

10 Proof of Theorem 2

Let $n \rightsquigarrow w$ along $y_0 \to \cdots \to y_s = w$. We show $y_{s-j} = 2^j w$ by downward induction. The only predecessors of x are $2x$ (always) and $(x-1)/p$ (only if $x \equiv 1 \pmod p$). If $x = 2^j w$ then $x \bmod p \in wH$, and $1 \in wH$ would give $w \in H$, a contradiction; so the odd predecessor does not occur, and the only predecessor is $2^{j+1}w$. Hence $n = 2^s w$. Every $n \equiv 0 \pmod p$ is excluded since $2^k w$ is a unit mod p. □

11 Remarks on scope

1. The bound is an upper bound on the least witness; it is not claimed sharp.
2. Primitivity of 2 is never used. The hypothesis on w enters only through Lemma 9 and Theorem 2.
3. The Wieferich case $p^2 \mid 2^{p-1} - 1$ is not treated here. A claimed failure of coverage at $p = 1093$ with $w = 1$ comes from outside this paper and is not proved or used here.
4. The restriction $N \geq 2$ is used only in Lemma 13.